SEMESTER – III							
S. NO.	COURSE CODE	COURSE NAME	COURSE TYPE#	PERIODS /WEEK		CREDITS	CATE GORY
				L-T-P	TCP*		
1	MA23C05	Probability and Statistics	T	3-1-0	4	4	HSMC
2	CS23301	Software Engineering	T	3-0-0	3	3	PCC
3	CS23302	Data Structures	LIT	3-0-4	7	5	PCC
4	CS23303	Digital System Design	LIT	3-0-4	7	5	ESC
5	CS23304	Java Programming	LIT	3-0-4	7	5	PCC
6		Skill Development Course – I	L	-	-	2	SDC
7	CS23U01	Standards – Computer Science & Engg.	T	1-0-0	1	1	UC
8	UC23U01	Universal Human Values	LIT	1-0-2	3	2	UC
					TOTAL CREDITS	27	

SEMESTER – IV							
S. NO.	COURSE CODE	COURSE NAME	COURSE TYPE#	PERIODS /WEEK		CREDITS	CATEG ORY
				L-T-P	TCP*		
1	MA23C03	Linear Algebra and Numerical Methods	T	3-1-0	4	4	HSMC
2	CS23401	Database Management Systems	LIT	3-0-4	7	5	PCC
3	CS23402	Computer Architecture	LIT	3-0-2	5	4	PCC
4	CS23403	Full Stack Technologies	LIT	2-0-4	6	4	PCC
5	CS23404	Design and Analysis of Algorithms	T	3-0-0	3	3	PCC
6		Skill Development Course - II	-	-	-	2	SDC
7		Audit Course II	-	-	-	-	UC
8		Industry Oriented Course I	-	-	-	1	SDC
					TOTAL CREDITS	23	

MA23C03	LINEAR ALGEBRA AND NUMERICAL METHODS	L	T	P	C
		3	1	0	4

OBJECTIVES:

- To understand Vector spaces and its basis and dimension.
- To understand the linear maps between vector spaces and their matrix representations.
- To understand the diagonalizatition of a real symmetric matrix.
- To understand Inner product spaces and its projections.
- To understand numerical techniques for solving linear systems, eigenvalue problems and generalized inverses.

UNIT I VECTORSPACES 9+3

Vector Spaces – Subspaces – Linear Combinations - Linear Span – Linear Dependence - Linear Independence – Bases and Dimensions.

UNIT II LINEAR TRANSFORMATIONS 9+3

Linear Transformation – Null Space, Range Space - Dimension Theorem - Matrix representation of Linear Transformation – Eigenvalues and Eigenvectors of Linear Transformation – Diagonalization of Linear Transformation – Application of Diagonalization in Linear System of Differential Equations.

UNIT III INNER PRODUCT SPACES 9+3

Inner Products and Norms - Inner Product Spaces - Orthogonal Vectors – Gram Schmidt Orthogonalization Process – Orthogonal Complement – Least Square Approximations.

UNIT IV NUMERICAL SOLUTION OF LINEAR SYSTEM OF EQUATIONS 9+3

Solution of Linear System of Equations – Direct Methods: Gauss Elimination Method – Pivoting, Gauss Jordan Method, LU Decomposition Method and Cholesky Decomposition Method - Iterative Methods: Gauss-Jacobi Method, Gauss-Seidel Method and SOR Method.

UNIT V NUMERICAL SOLUTION OF EIGENVALUE PROBLEMS AND GENERALISED INVERSES 9+3

Eigen Value Problems: Power Method – Inverse Power Method – Jacobi's Rotation Method - QR Decomposition - Singular Value Decomposition Method.

TOTAL: 60 PERIODS

Laboratory based exercises / assignments / assessments will be given to students from the content of the course wherever applicable.

Branch specific / General Engineering applications based on the content of each units will be introduced to students wherever possible.

Suggested Laboratory based exercises / assignments / assessments :

1. Linear independence/dependence of vectors
2. Computation of eigenvalues and eigenvectors
3. Diagonalization of Linear Transformation
4. Gram Schmidt Orthogonalization Process
5. Solution of algebraic and transcendental equations
6. Matrix Decomposition methods (LU / Cholesky Decomposition)
7. Iterative methods of Gauss-Jacobi and Gauss-Seidel
8. Matrix Inversion by Gauss-Jordan method

9. Eigen values of a matrix by Power method and by Jacobi's method
10. QR decomposition method
11. Singular Value Decomposition Method

OUTCOMES:

CO1: Solve system of linear equations using matrix operations and vector spaces using Algebraic methods.
CO2: Understand the linear maps between vector spaces and its utilities.
CO3: Apply the concept of inner product of spaces in solving problems.
CO4: Understand the common numerical methods and how they are used to obtain approximate solutions
CO5: Analyse and evaluate the accuracy of common numerical methods.

TEXT BOOKS:

1. Faires, J.D. and Burden, R., "Numerical Methods", Brooks/Cole (Thomson Publications), Fourth Edition, New Delhi, 2012.
2. Friedberg, S.H., Insel, A.J. and Spence, E., "Linear Algebra", Pearson Education, Fifth Edition, New Delhi, 2018.
3. Williams, G, "Linear Algebra with Applications", Jones & Bartlett Learning, First Indian Edition, New Delhi, 2019.

REFERENCES:

1. Bernard Kolman, David R. Hill, "Introductory Linear Algebra", Pearson Education, First Reprint, New Delhi, 2010.
2. Gerald, C.F, and Wheatley, P.O., "Applied Numerical Analysis", Pearson Education, Seventh Edition, New Delhi, 2004.
3. Kumaresan, S., "Linear Algebra – A geometric approach", Prentice – Hall of India, Reprint, New Delhi, 2010.
4. Richard Branson, "Matrix Operations", Schaum's outline series, Mc Graw Hill, New York, 1989.
5. Strang, G., "Linear Algebra and its applications", Cengage Learning, New Delhi, 2005.

CO – PO Mapping:

Course Outcomes	PROGRAMME OUTCOMES											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	P12
CO 1 :	3	3	2	3	1	2	1	1	1	1	1	3
CO 2 :	3	3	2	3	1	2	1	1	1	1	1	3
CO 3 :	3	3	2	3	1	2	1	1	1	1	1	3
CO 4 :	3	3	2	3	1	2	1	1	1	1	1	3
CO 5 :	3	3	2	3	1	2	1	1	1	1	1	3

CS23401 DATABASE MANAGEMENT SYSTEMS

L	T	P	C
3	0	4	5

UNIT – I INTRODUCTION TO DATABASE SYSTEMS 9L, 8P

Introduction to Databases - File System Vs Database System - Data Models - Schemas and Instances - DBMS Architecture - Centralized - Client Server - Database Applications - ER Models – ER to Relational Mapping

PRACTICALS:

1. ER Diagram

UNIT – II RELATIONAL MODELS 10L, 24P

Relational Model - Constraints - Keys - Dependencies - Relational Algebra - Unary, Binary, Set and Extended Relational Algebra operations – SQL- Data Definition - Data Manipulation and Retrieval Queries - Nested Queries – Joins - Views-Cursors -Procedures – Cursors-Procedures - Triggers - Embedded and Dynamic SQL

PRACTICALS:

DDL commands:

1. Creation of tables with appropriate integrity constraints
2. Usage of alter, drop commands

DML commands:

3. Data Insertion, updation, and deletion with tables
4. Data retrieval using
   - Simple SQL
   - Nested Queries
5. Different types of Joins
6. PL/SQL: Functions, Procedures and Triggers

UNIT – III RELATIONAL DATABASE DESIGN 7L, 4P

Database Design - Functional Dependencies – Normalization - 1 NF - 2 NF - 3 NF - BCNF - Multivalued Dependency (4 NF) - Join Dependency (PJNF)

PRACTICALS:

1. Aggregation operators-Grouping and ordering

UNIT – IV TRANSACTIONS AND RECOVERY 10L, 12P

Transaction processing concepts - Need for concurrency control and recovery- ACID Properties - Recoverability – Serializability - Concurrency Control - Two phase locking Techniques – Timestamp based protocol – Graph based protocol - Deadlock handling – Log based recovery – Two Phase Commit Protocol

PRACTICALS:

Create View tables

1. DCL Commands: Grant and Revoke
2. TCL Commands: Save point, Commit, Rollback

UNIT – V QUERY PROCESSING AND ADVANCED DATABASES 9L, 12P

Indexing and Hashing Techniques - Query Processing and Optimization - Sorting and Joins – Database Tuning – Introduction to Spatial and Temporal Databases – OO Databases – NoSQL

PRACTICALS:

1. Implementation of suitable front end for querying and displaying the results

TOTAL: 45L + 60P = 105 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

1. Model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model
2. Formulate solutions to a broad range of query problems using relational algebra/ SQL
3. Apply normalization theory to normalize the relations in RDBMS to avoid redundancy and anomalies.
4. Manage concurrent transactions and their consequences, and analyze the use of triggers, functions, and procedures in a realistic database application.
5. Understand database storage structures and access techniques

REFERENCES:

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Seventh Edition, Tata McGraw Hill, 2019.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Seventh Edition, Pearson/Addison - Wesley, 2016.
3. C.J. Date, A. Kannan and S. Swamynathan, "An Introduction to Database Systems", Pearson Education, Eighth Edition, 2006.
4. Raghu Ramakrishnan and Johannes Gehrke, "Database Management Systems", Third Edition, McGraw Hill, 2014.
5. Andreas Meier, Michael Kaufmann, "SQL & NoSQL Databases: Models, Languages, Consistency Options and Architectures for Big Data Management", 1st ed. 2019 Edition.
6. Narain Gehani and Melliyal Annamalai, "The Database Book: Principles and Practice Using the Oracle Database System", Universities Press, 2012.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	3	1	-	2	-	1	3	3	3	2
CO2	3	3	3	3	3	3	1	-	2	-	1	2	3	3	2
CO3	3	3	3	3	2	3	-	-	3	-	1	3	3	3	2
CO4	3	3	3	3	3	3	-	-	1	-	1	2	3	3	2
CO5	3	3	3	2	3	3	-	-	2	-	1	3	3	3	2

1-low, 2-medium, 3-high, '-"- no correlation

CS23402 COMPUTER ARCHITECTURE

L	T	P	C
3	0	2	4

UNIT – I INSTRUCTION SET ARCHITECTURE 9L, 10P

Introduction - Classes of computer systems - Performance - Amdahl's law - The Power wall - Switch from uniprocessors to multiprocessors - Benchmarks. Hardware Software Interface - ISA - Operations of the computer hardware - Operands - Representing instructions - Instructions for making decisions - Supporting procedures in computer hardware. Addressing modes - Translating and starting a program - Arrays versus pointers - MIPS instruction formats - Assembly language programming.

PRACTICALS:

1. Study of an existing standard architectural simulator.
2. Study of addressing modes with examples, tracing the execution sequences, identifying the timing constraints.
3. Study of the ISA supported by the architectural simulator and running simple programs on the simulator.

UNIT – II ARITHMETIC FOR COMPUTERS 9L

Integer arithmetic - Binary Parallel adder – Carry Look-ahead Adder - Carry save adder - Binary multiplier - Booth's multiplier - Bit-pair recoding - Binary division. Floating point arithmetic-Representation - Arithmetic operations on floating point numbers - Parallelism and computer arithmetic.

UNIT – III PROCESSOR DESIGN 9L, 10P

Datapath design - Implementation of the basic MIPS ISA - Building the datapath - A simple implementation scheme - Drawbacks. Instruction Level Parallelism - Pipelining - Performance - Pipeline hazards - Pipelined datapath and control - Handling data hazards and control hazards – Exceptions.

PRACTICALS:

1. Analysing the datapath on the standard simulator.
2. Study of the pipelined implementation and analysis of various hazards on the standard simulator

UNIT – IV ADVANCED ILP 9L

Advanced ILP – Dynamic branch prediction – Correlating predictors – Tournament predictors. Dynamic scheduling – Tomasulo's algorithm – Speculation. Multiple issue processors – Static and dynamic. Limitations of ILP - Multithreading – SMT and CMP Architectures – The Multicore era.

UNIT – V MEMORY AND INPUT/OUTPUT 9L, 10P

Need for a hierarchical memory system - The basics of caches - Measuring and improving cache performance. Virtual memory - Paging and segmentation - TLB - Implementing protection with virtual memory. Associative memories, Introduction to virtual machines.

Storage and I/O - Dependability, reliability and availability –Types of storage. Connecting processors, memory and I/O devices - Interfacing I/O devices to the processor, memory and the operating system, Interrupts, DMA, RAID.

PRACTICALS:

1. Implement a simple functional model of a set-associative cache in C/C++. Study hit/miss rates for various access patterns. Experiment with different replacement policies.
2. Writing simple programs to study the behaviour of the memory hierarchy.
3. Analyzing the performance of the memory hierarchy by varying different parameters.

TOTAL: 45L + 30P = 75 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

1. Evaluate the performance of computer systems and write simple MIPS assembly language programs
2. Design a simple instruction execution unit
3. Point out the hazards present in a pipeline and suggest remedies
4. Show how ILP is exploited while executing a sequence of instructions
5. Discuss the working of an architectural simulator and modify some features of it
6. Critically analyse the various characteristics of the hierarchical memory and I/O devices and their interface to the processor

REFERENCES:

1. David A. Patterson and John L. Hennessy, “Computer Organization and Design: The Hardware/Software Interface”, Sixth Edition, Morgan Kaufmann / Elsevier, 2020.
2. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, “Computer Organization and Embedded Systems”, Sixth Edition, Tata McGraw Hill, 2012.
3. John L. Hennessey and David A. Patterson, “Computer Architecture – A Quantitative Approach”, Morgan Kaufmann / Elsevier, 6th edition, 2019.
4. William Stallings, “Computer Organization and Architecture – Designing for Performance”, Tenth Edition, Pearson Education, 2016.
5. John P. Hayes, “Computer Architecture and Organization”, Third Edition, Tata McGraw Hill, 2017.
6. V.P. Heuring, H.F. Jordan, “Computer Systems Design and Architecture”, Second Edition, Pearson Education, 2004.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	1	1	1	-	-	-	-	2	3	3	3
CO2	3	3	3	2	1	1	1	-	1	-	-	2	3	3	3
CO3	3	3	3	2	1	1	1	-	-	-	-	2	3	3	3
CO4	3	3	3	2	1	1	1	-	1	-	-	2	3	3	3
CO5	3	3	3	2	1	1	1	-	1	-	-	2	3	3	3
CO6	3	3	3	2	1	1	1	-		-	-	2	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

CS23403	FULL STACK TECHNOLOGIES	L	T	P	C
		2	0	4	4

UNIT – I OVERVIEW OF FULL STACK 6L, 12P

Understanding the Basic Web Development Framework - Browser – JavaScript – functions, arrays, objects, strings, Webserver — Backend Services – MVC Architecture - different stacks – The MEAN / MERN stacks

PRACTICALS:

1. Experiment with JavaScript – functions, arrays, strings, objects

UNIT – II NODE.JS AND EXPRESS 6L, 12P

Basics of Node JS – Installation – Working with Node packages – Using Node package manager – Creating a simple Node.js application – Using Events – Listeners –Timers - Callbacks – Handling Data I/O – Implementing HTTP services in Node.js

PRACTICALS:

1. Install Express with Node.js, exploring modules
2. HTTP servers and Clients

UNIT – III FRONT-END DEVELOPMENT 6L, 12P

Angular – CLI – Typescript - Using Request and Response objects - modules, components, templates, metadata, data binding, directives, services, dependency injection. React - Virtual DOM, components, props, JSX, Events, conditionals, lists, forms, Routing, Hooks.

PRACTICALS:

1. Using Angular modules, directives, components, templates, data binding, controllers
2. REACT – components, JSX, props, Rendering, forms

UNIT – IV MONGODB 6L, 12P

Understanding NoSQL and MongoDB – Building MongoDB Environment – User accounts – Access control – Administering databases – Managing collections – Connecting to MongoDB from Node.js – simple applications

PRACTICALS:

1. NoSQL with MongoDB – setting up a document DB, connecting to MongoDB, CRUD operations

UNIT – V BUILDING WEB APPLICATIONS AND SERVICES 6L, 12P

Single Page Applications, Web Services and APIs, Microservices – architecture, Developing Single Page Applications, Building RESTful APIs, Building web services and micro services, Deployment, TLS Certificate.

PRACTICALS:

1. Developing single Page Applications, RESTful APIs, Web Services, Microservices

TOTAL: 30L + 60P = 90 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

1. Use JavaScript and its libraries for building front-end of web applications.
2. Use Node.js for application development
3. Use the features of Angular and React for developing Single Page Applications
4. Develop applications with MongoDB
5. Develop Web applications, APIs and Services using full stack

REFERENCES:

1. Brad Dayley, Brendan Dayley, Caleb Dayley, 'Node.js, MongoDB and Angular Web Development', Addison-Wesley, Second Edition, 2018.
2. Vasan Subramanian, 'Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node', Second Edition, Apress, 2019.
3. David Herron, Node.js Web Development, Packt Publishing Limited, 5th edition, 2020.
4. David Flanagan, Javascript The Definitive Guide, Oreilly, 7th Edition, 2020.
5. Nate Murray, Felipe Coury, Ari Lerner, Carlos Taborda, Ng-book the complete Book on Angular, Fullstack.io, 2020.
6. Greg Lim, Beginning MEAN Stack (MongoDB, Express, Angular, Node.js), Independently Published, 1st Edition, 2021.
7. Greg Lim, Beginning Node.js, Express & MongoDB Development, Independently Published, 2020.
8. https://nodejs.org/en
9. https://angular.io
10. https://react.dev

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO4	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

CS23404	DESIGN AND ANALYSIS OF ALGORITHMS	L	T	P	C
		3	0	0	3

UNIT – I FUNDAMENTALS 6L

Introduction – Asymptotic Notations – Recurrences – Substitution Method - Recurrence Tree Method - Master Method - Probabilistic Analysis and Randomized Algorithms - Best, Worst and Average Case Analysis of Linear Search, Binary Search and Insertion Sort.

UNIT – II DIVIDE & CONQUER STRATEGY 15L

Maximum Subarray – Strassen's Matrix Multiplication - Analysis of Quick Sort, Merge Sort – Quick Sort Randomized Version – Sorting in Linear Time - Lower Bounds for Sorting - Selection in Expected Linear Time - Selection in Worst case Linear Time.

UNIT – III DYNAMIC PROGRAMMING AND GREEDY STRATEGIES 8L

Elements of Dynamic Programming - Rod cutting - Matrix Chain Multiplication – Longest Common Subsequence – Elements of Greedy Strategy - Huffman Code - Single source shortest path.

UNIT – IV OTHER DESIGN STRATEGIES AND STRING MATCHING 8L

Backtracking: N-Queens – Branch & Bound: Travelling Salesperson – String Matching: Naïve, KMP

UNIT – V NP PROBLEMS 8L

NP-Completeness – Polynomial Time Verification – Theory of Reducibility - Circuit Satisfiability – NP - Completeness Proofs – NP Complete Problems: Vertex Cover, Hamiltonian Cycle and Traveling Salesman Problems – Approximation Algorithms – Approximation Algorithms to Vertex - Cover and Traveling Salesman Problems.

TOTAL = 45 PERIODS

COURSE OUTCOMES:

Upon completion of the course, the students will be able to

1. Have a strong foundation for algorithm study.
2. Analyze the asymptotic performance of algorithms.
3. Apply important algorithmic design paradigms and methods of analysis.
4. Demonstrate familiarity with string matching algorithms.
5. Prove the hardness and to find alternatives for such problems.

REFERENCES:

1. Thomas H Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, “Introduction to Algorithms”, Third Edition, Prentice Hall, 2010.
2. Ellis Horowitz, Sartaj Sahni and Senguthevar Rajasekaran, Fundamentals of Computer Algorithms, Second Edition, University Press, 2007
3. Gilles Brassard and Paul Bratley, Fundamentals of Algorithmics, Prentice-Hall, 1996.
4. Kenneth A. Berman and Jerome L Paul, “Algorithms”, Cengage Learning India, 2010
5. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, fourth edition, Pearson, 2014.
6. Dasgupta S, Papadimitriou H C and Vazirani U V , Algorithms, 2006.
7. Anany Levitin, “Introduction to the Design and Analysis of Algorithms”,Third Edition, Pearson, 2012.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	-	3	-	-	-	-	-	-	3	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	3	3	3	3
CO3	3	3	3	3	3	1	-	-	-	-	-	3	3	3	3
CO4	3	3	3	3	3	1	-	-	-	-	-	3	3	3	3
CO5	3	3	2	3	3	1	-	-	-	-	-	3	3	3	3

1-low, 2-medium, 3-high, ‘-“- no correlation